INCH-POUND

MS18100E
29 June 2007
SUPERSEDING
MS18100D
25 July 1983

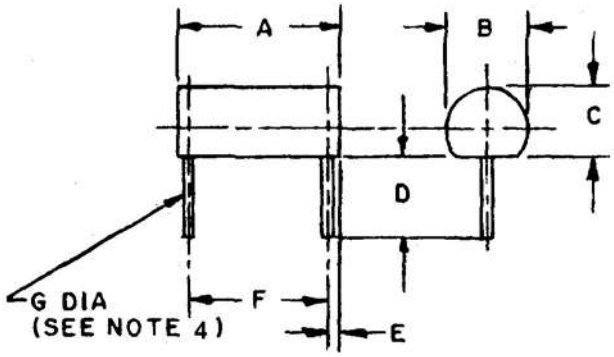
MILITARY SPECIFICATION SHEET

COILS, RADIO FREQUENCY, MOLDED, FIXED, SUBMINIATURE, (PHENOLIC CORE), TYPES LT4K200 TO LT4K208 INCL. LT4K220 and LT4K221

Inactive for new design after 25 July 1983.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.



Ltr	Dimensions in inches with metric equivalents (mm) in parentheses			
	Dash numbers 1 through 9		Dash numbers 21 and 22	
	Minimum	Maximum	Minimum	Maximum
A	.694 (17.63)	.756 (19.20)	.594 (15.09)	.656 (16.66)
B	.204 (5.18)	.266 (6.76)	.251 (6.38)	.313 (7.95)
C	.189 (4.80)	.251 (6.38)	.234 (5.94)	.296 (7.52)
D	.250 (6.35)	.374 (9.50)	.250 (6.35)	.374 (9.50)
E	.051 (1.30)	.071 (1.80)	.051 (1.30)	.071 (1.80)
F	.590 (14.99)	.610 (15.49)	.490 (12.45)	.510 (12.95)
G	.023 (0.58)	.027 (0.69)	.023 (0.58)	.027 (0.69)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents, in parentheses, are given for general information only.
3. These coils are intended to be mounted by the body.
4. Solderable/weldable lead wire, AWG number 22.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Style: LT4.

Grade: 1.
Class: B.

Weight: 1.5 grams, maximum. (Dash numbers 1 through 9)
Weight: 1.8 grams, maximum. (Dash numbers 21 and 22)

Operating temperature range: -55°C to +125°C.

Ambient temperature: 90°C maximum.

Temperature rise: 35°C maximum.

Terminal pull: 5 pounds minimum.

Altitude: 70,000 feet.

Shock, specified pulse: Method 213 of MIL-STD-202, test condition I, is applicable.

Dielectric withstanding voltage:

At sea level: Method 301 of MIL-STD-202, test voltage 1,000 V rms for a minimum of 60 seconds.

At reduced barometric pressure: Method 105 of MIL-STD-202, test condition C, test voltage 200 V rms for a minimum of 60 seconds.

Electrical characteristics: See table I and table II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I.

Part or Identifying Number (PIN): MS18100- (dash number from table I).

VERIFICATION:

Quality assurance provisions:

Qualification Inspection: Not applicable for this standard.

Conformance Inspection: Group A and group B tests of MIL-PRF-15305 shall be applicable.

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TABLE I. Electrical characteristics (initial).

Dash number 1/ 2/	Type designation	Superseded MS PIN	Inductance (μH)	Q (min)	Test frequency (MHz)	SRF min (MHz)	DC resistance (ohms)	Rated dc current (mA)
-1	LT4K200	MS16226-1	0.47 ±20%	50	25	250	.06	1,800
-2	LT4K201	MS16226-2	0.68 ±20%	50	25	215	.10	1,400
-3	LT4K202	MS16226-3	1.00 ±10%	50	25	180	.15	1,150
-4	LT4K203	MS16226-4	1.50 ±10%	40	7.9	145	.28	850
-5	LT4K204	MS16226-5	2.20 ±10%	40	7.9	120	.50	625
-6	LT4K205	MS16226-6	2.70 ±10%	40	7.9	108	.70	525
-7	LT4K206	MS16226-7	3.30 ±10%	40	7.9	95	1.40	370
-8	LT4K207	MS16226-8	3.90 ±10%	40	7.9	90	1.50	350
-9	LT4K208	MS16226-9	4.70 ±10%	40	7.9	80	1.80	325
-21	LT4K220	MS16226-21	47 ±10%	15	2.5	8.5	5.9	200
-22	LT4K221	MS16226-22	56 ±10%	15	2.5	8.3	6.4	175

1/ The dash number added to MS military standard number constitutes the MS PIN; for example MS18100-1.
2/ The former MS part numbers MS18100-10 through MS18100-20 and MS18100-23 through MS18100-42 have been superseded as follows:
MS18100-10 through MS18100-20 by MS14040-1 through MS14040-11, respectively,
MS18100-23 through MS18100-27 by MS14041-1 through MS14041-5, respectively,
MS18100-28 through MS18100-32 by MS14042-1 through MS14042-5, respectively,
MS18100-33 through MS18100-37 by MS14043-1 through MS14043-5, respectively,
MS18100-38 through MS18100-42 by MS14044-1 through MS14044-5, respectively,

TABLE II. Electrical characteristics (final). 1/

Inspection group	Allowable variation from Initial measurement		Allowable percent from specified minimum value in electrical characteristics (initial) table	
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±2	---	---	-10
Group III	±5	±(3% +.001 ohm)	-8	-10
Group IV 2/	±5	±(2% +.001 ohm)	-10	-15
Conformance inspection group C				
Subgroup I	±2	---	---	-10
Subgroup II 2/	±5	±(2% +.001 ohm)	-10	-15
Subgroup III	±5	±(3% +.001 ohm)	-8	-10

1/ Test fixture allowance of +.01 μH shall be added to all change in inductance limits ±(_ percent +.01 μH).
2/ The polarizing voltage during the moisture resistance tests Is applied with the positive lead connected to the coil terminals tied together, and the negative lead connected to the metal strap.

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Referenced documents. In addition to MIL-PRF-15305, this document references MIL-STD-202.

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA - CC

Review activities:
Army – AR, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19

Preparing activity:
DLA – CC

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